

Can a Vision–Language Model Improve the Grouping of 2D Masks into 3D Object Instances?

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Motivation



Geometry groups **2D masks** consistently across views, but lacks the **semantic understanding** needed to resolve ambiguous object associations.

Research question: Can semantic evidence improve geometric mask grouping without requiring pairwise VLM inference?

Three Approaches We Compare

1 MaskClustering

Geometric baseline

Mask backprojection → derivation of observer frames → view consensus → graph clustering → 3D instances.
Efficient, but limited by geometric evidence.

2 Fine-tuned Qwen3-VL

Semantic edge scorer

Two frames with outlined masks each → $P(\text{same instance})$ → graph clustering.

3 Fusion MLP (ours)

Semantic–geometric fusion

Semantic + geometric features → $P(\text{same instance})$ → graph clustering.

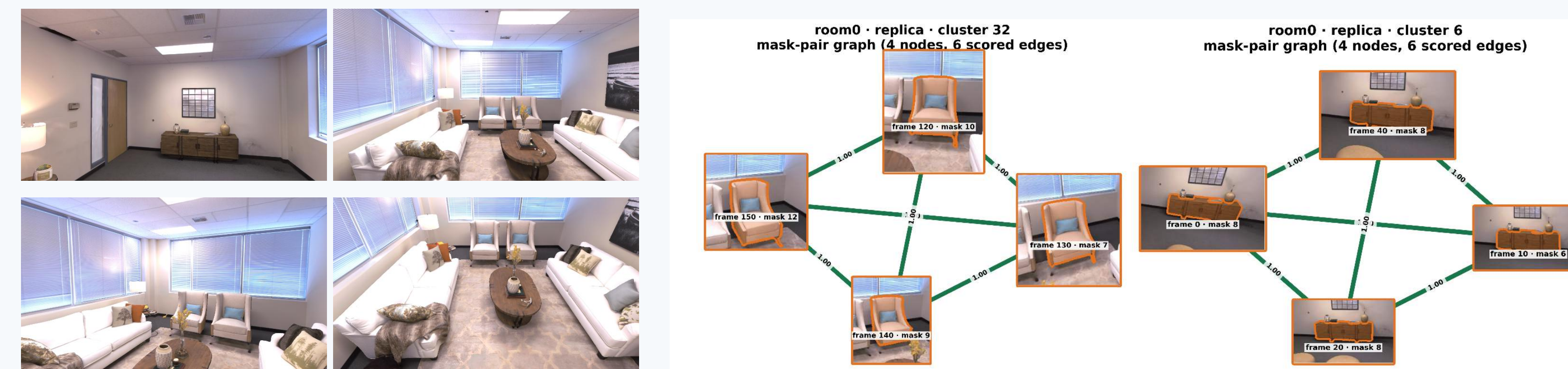
Takeaway & Limitations

KEY TAKEAWAY

On Replica, geometry carries most of the signal: learned fusion slightly beats the baseline, while the fine-tuned VLM alone collapses objects together. LoRA likely hurts due to too little training data.

- **Candidate-selection bottleneck:** Upstream filtering may discard valid mask pairs before semantic–geometric fusion.
- **Limited fine-tuning data:** The small training set limits robust adaptation and generalization of Qwen.
- **Computational scalability:** VLM inference remains too expensive for evaluating all possible mask pairs.

Milestone 1 · Reproducing MaskClustering



Milestone 2 · Choosing the 2D Mask Generator



Milestone 3 · Using Qwen3-VL and Fine-tuned Qwen3-VL

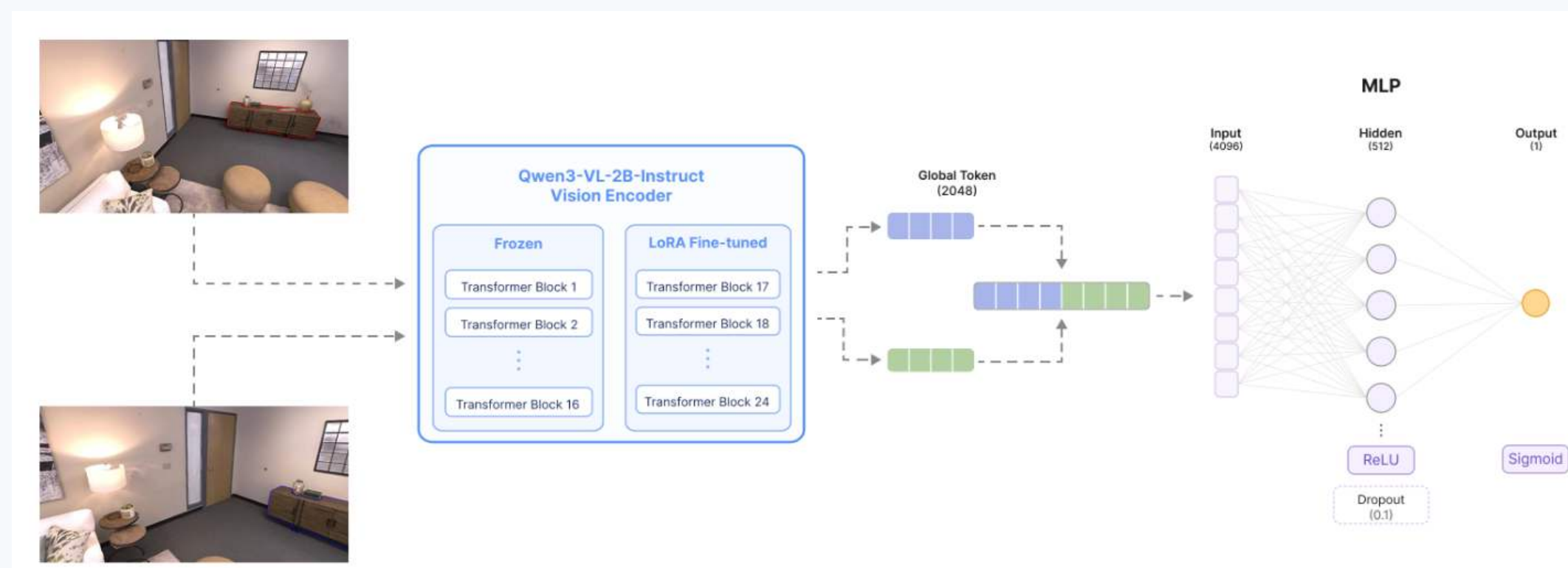
High-recall candidate selection

Qwen inference on all $\mathcal{O}(N^2)$ pairs is too slow. We therefore retain a pair if **any** cheap criterion passes:

$$\begin{aligned} &\text{DepthIoU} \geq \tau \\ &\text{OR directional overlap} \geq \tau \\ &\text{OR view consensus} \geq 0.60 \end{aligned}$$

The relaxed OR rule aims to preserve positive pairs while reducing expensive VLM calls.

Fine-Tuned QWEN3-VL architecture



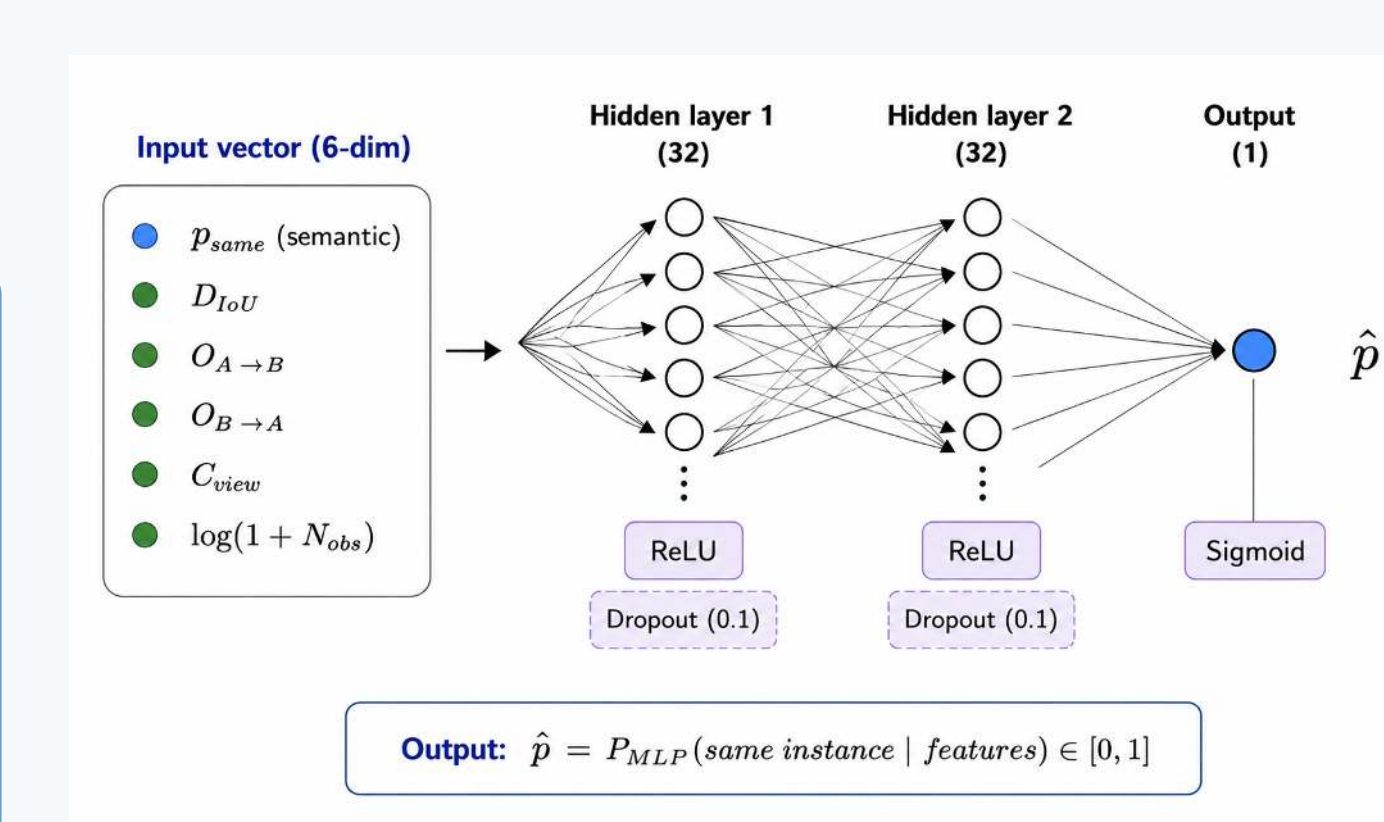
Milestone 4 · Efficient Semantic-Geometric Fusion

1. Fusion features

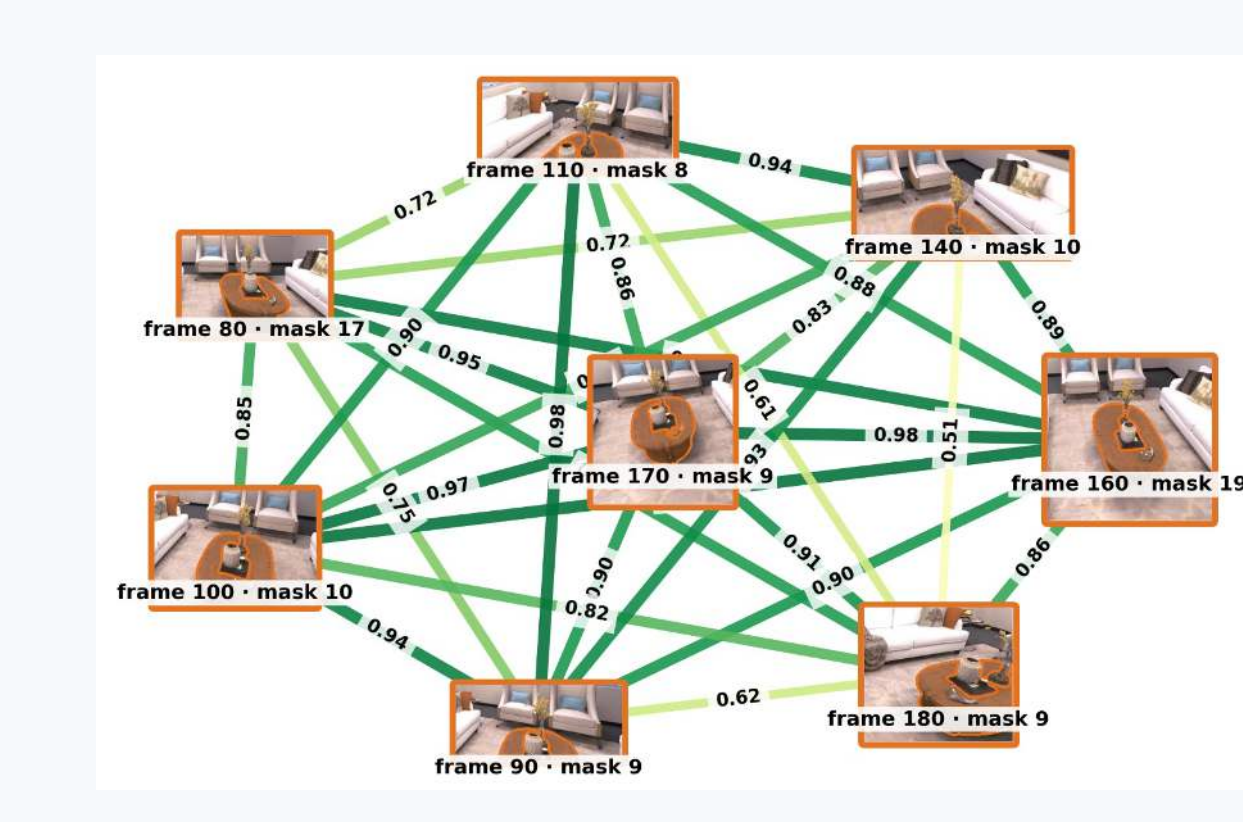
Each retained mask pair is represented by one semantic and five geometric features.

Semantic p_{Qwen}
Geometric C_{view}
 $\log(1 + N_{\text{obs}})$
DepthIoU
 $O_{A \rightarrow B}$
 $O_{B \rightarrow A}$

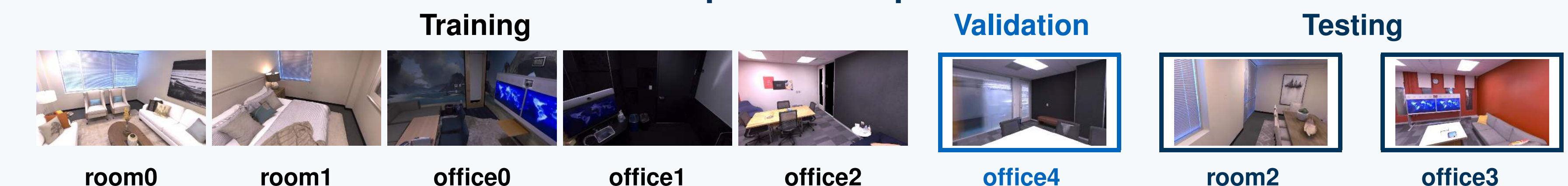
2. Fusion MLP edge scorer



3. MaskClustering integration



Replica data split



Results · Replica room2

Edge scorer	AP	AP@50	AP@25
View consensus (baseline)	0.082	0.189	0.423
Qwen zero-shot	0.019	0.032	0.105
Qwen fine-tuned (LoRA)	0.018	0.033	0.173
Average (Qwen + consensus)	0.075	0.191	0.430
Fusion MLP (geo + Qwen)	0.083	0.205	0.441

Leak-free test scene (Qwen fine-tuned on office0/office1/room0). The Fusion MLP slightly beats the geometric baseline; simple averaging matches it and Qwen-only stays near zero. All variants share the same clustering and CropFormer masks.

Mask backbone (M2)	AP	AP@50	AP@25
CropFormer	0.149	0.305	0.489
SAM3 (conf 0.25)	0.123	0.224	0.370
SAM1 (ViT-H)	0.061	0.107	0.222

Qwen Model Fine-Tuning

Model	room2 (test)			office3 (test)		
	AP	AP@50	AP@25	AP	AP@50	AP@25
Qwen zero-shot	0.019	0.032	0.105	0.013	0.046	0.155
Fine-tuned ($\tau = 0.50$)	0.010	0.010	0.091	0.016	0.056	0.113
Fine-tuned ($\tau = 0.68$)	0.018	0.032	0.191	0.031	0.099	0.173

Observation: Fine-tuning yields only small and inconsistent gains over zero-shot Qwen. The available training set is likely too limited for robust model adaptation.

Qualitative Results

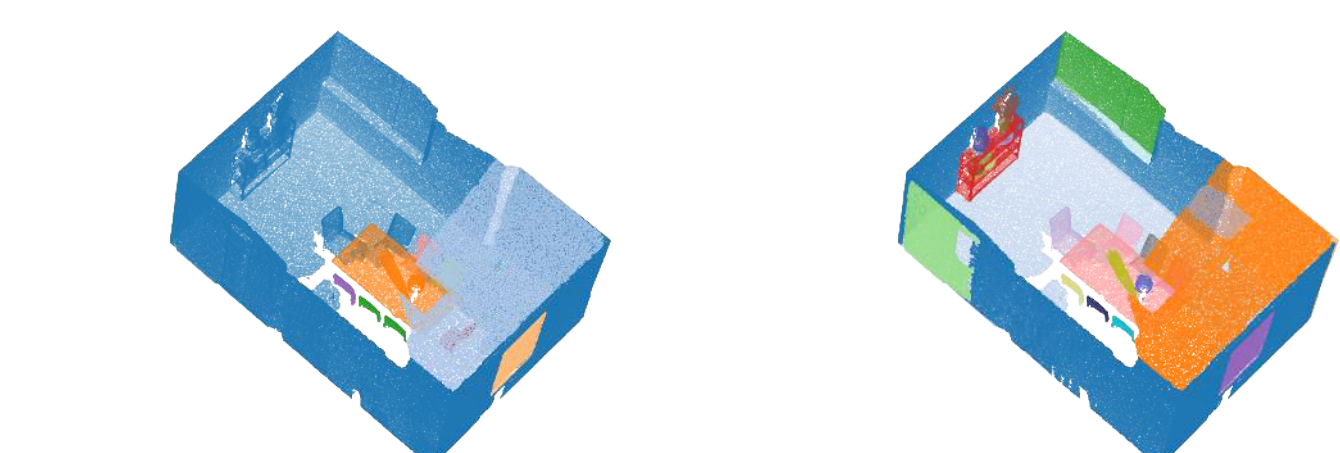
MaskClustering

Qwen zero-shot



Qwen fine-tuned

Fusion MLP



Baseline (zoom)

Fusion MLP (zoom)

